

EXP-02: EITT on US Monthly Energy Compositions

9-Carrier Electricity Generation Simplex | $D = 9$ | $N = 300$

Higgins Iterative Validation Protocol (HIVP) Chain: EXP-01 → EXP-02

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Abstract. This journal documents the second experiment in the EITT validation chain. EXP-02 applies the Entropy-Invariant Time Transformer to US monthly electricity generation data across 10 states organized into interior (4), bridge (2), and boundary (4) cohorts. The 9-carrier simplex (Coal, Gas, Nuclear, Hydro, Solar, Wind, Bioenergy, Other Fossil, Other Renewables) tests EITT at $D = 9$ with 300 monthly observations decimated at $M = 2, 3, 4, 6, 12$. All four interior states pass at the 1% threshold. A new finding emerges: the geometric-arithmetic gap follows a linear-with-saturation pattern in M , providing a built-in contamination meter that measures near-vertex stress without modifying the core EITT observable. This linear contamination tuner is formalized as instrument F17.

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1. Experiment Design and Data

EXP-02 extends the EITT validation chain from EXP-01's gold/silver 2-simplex ($D = 2$, $N = 624$) to real-world energy compositions at $D = 9$. The source is EMBER's US monthly electricity generation dataset (CC BY 4.0), filtered to the 9-carrier fuel breakdown in percentage shares. Each state provides $N = 300$ monthly observations from January 2001 to December 2025.

Cohort selection follows the EXP-02 Formal Ranked Selection Memo, which defines exact gates for interior, bridge, and boundary cases based on mean entropy, dominant-share concentration, zero-share rate, and compositional velocity. Interior cases are diversified states far from the simplex boundary. Bridge cases sit near the 1% threshold. Boundary cases are concentrated, sparse, or volatile compositions expected to stress or break EITT.

Parameter	Value
Source	us_monthly_full_release_long_format.csv (EMBER)
D (components)	9 fuel carriers
N (observations)	300 months per state (2001-01 to 2025-12)
M tested	2, 3, 4, 6, 12
Pass threshold	1% relative entropy change at all M
Zero policy	Multiplicative replacement, $\epsilon = 10^{-6}$
Mapping	$f(r) = r$ (identity), close to simplex
Interior cohort	California, Minnesota, Texas, Wisconsin
Bridge cohort	Pennsylvania, North Carolina
Boundary cohort	Rhode Island, West Virginia, Wyoming, Delaware

Table 1. EXP-02 experiment design parameters.

2. Cohort Comparison Summary

The cohort comparison table below presents the primary EITT output alongside companion diagnostics for all 10 entities. The interior cohort passes unanimously. Pennsylvania (bridge) fails at $M = 12$ only. All boundary states fail at multiple M values.

State	Band	N	$H_{\text{bar}}(1)$	max rel%	RMS rel%	Gamma	delta min	mean v	eta	Verdict
California	interior	300	1.4850	0.829%	0.4768%	1318934	1.0e-06	0.828	0.0662	PASS
Minnesota	interior	300	1.3638	0.490%	0.3249%	1822512	1.0e-06	1.044	0.0464	PASS
Texas	interior	300	1.2193	0.208%	0.1199%	1737693	1.0e-06	0.998	0.0409	PASS
Wisconsin	interior	300	1.1813	0.134%	0.0820%	1388157	1.0e-06	0.788	0.0456	PASS
Pennsylvania	bridge	300	1.1235	1.092%	0.6674%	1635993	1.0e-06	1.364	0.0265	FAIL
North Carolina	bridge	300	1.2221	0.213%	0.1260%	2606342	1.0e-06	1.280	0.0347	PASS
Rhode Island	boundary	300	0.2785	8.784%	6.2649%	4817095	1.0e-06	2.065	0.0218	FAIL
West Virginia	boundary	300	0.2885	5.283%	3.4043%	1513491	1.0e-06	1.363	0.0321	FAIL
Wyoming	boundary	300	0.5112	2.824%	1.7752%	1078225	1.0e-06	0.812	0.0337	FAIL
Delaware	boundary	300	0.6948	6.441%	4.6053%	5096980	1.0e-06	2.580	0.0266	FAIL

Table 2. Cohort comparison summary. Gamma is computed at $M = 2$. All delta_min values are at the zero-replacement floor (10^{-6}).

3. Interior Cohort: Per-M Decimation Tables

All four interior states pass at every tested M. The maximum relative entropy change is California at M = 12 (0.829%), with 17 basis points of headroom below the 1% threshold. Texas is the strongest performer (max 0.208%). SNR remains below 2 at all stations, meaning all EITT deltas are below the statistical noise floor — the same pattern observed at D = 2 in EXP-01.

California — $H_{\text{bar}}(1) = 1.484997$, $\sigma_A^2 = 2.0192$, verdict: PASS

M	blocks	delta_M	rel %	SE(M)	SNR	ETF	phase RMSE	pass
2	150	0.001072	0.072%	0.011138	0.0962	2.65e-10	0.000416	PASS
3	100	0.002046	0.138%	0.013641	0.1500	7.60e-10	0.000910	PASS
4	75	0.006721	0.453%	0.015752	0.4267	3.33e-09	0.001421	PASS
6	50	0.006964	0.469%	0.019292	0.3610	5.17e-09	0.003051	PASS
12	25	0.012316	0.829%	0.027282	0.4514	1.83e-08	0.001051	PASS

Minnesota — $H_{\text{bar}}(1) = 1.363835$, $\sigma_A^2 = 2.7901$, verdict: PASS

M	blocks	delta_M	rel %	SE(M)	SNR	ETF	phase RMSE	pass
2	150	0.001477	0.108%	0.020008	0.0738	2.65e-10	0.000099	PASS
3	100	0.002726	0.200%	0.024505	0.1113	7.33e-10	0.001036	PASS
4	75	0.004180	0.306%	0.028296	0.1477	1.50e-09	0.001262	PASS
6	50	0.005143	0.377%	0.034656	0.1484	2.76e-09	0.000264	PASS
12	25	0.006680	0.490%	0.049011	0.1363	7.18e-09	0.000347	PASS

Texas — $H_{\text{bar}}(1) = 1.219310$, $\sigma_A^2 = 2.6603$, verdict: PASS

M	blocks	delta_M	rel %	SE(M)	SNR	ETF	phase RMSE	pass
2	150	0.000132	0.011%	0.009514	0.0139	2.49e-11	0.000049	PASS
3	100	0.000677	0.056%	0.011652	0.0581	1.91e-10	0.000130	PASS
4	75	0.000782	0.064%	0.013455	0.0581	2.94e-10	0.000521	PASS
6	50	0.001778	0.146%	0.016479	0.1079	1.00e-09	0.000568	PASS
12	25	0.002537	0.208%	0.023304	0.1089	2.86e-09	0.000114	PASS

Wisconsin — $H_{\text{bar}}(1) = 1.181324$, $\sigma_A^2 = 2.1252$, verdict: PASS

M	blocks	delta_M	rel %	SE(M)	SNR	ETF	phase RMSE	pass
2	150	0.000073	0.006%	0.015335	0.0048	1.72e-11	0.000165	PASS
3	100	-0.000295	0.025%	0.018782	0.0157	1.04e-10	0.000197	PASS
4	75	-0.000620	0.052%	0.021688	0.0286	2.92e-10	0.000349	PASS
6	50	-0.001584	0.134%	0.026562	0.0596	1.12e-09	0.000577	PASS
12	25	-0.001307	0.111%	0.037564	0.0348	1.85e-09	0.000409	PASS

Table 3. Per-M decimation results for the interior cohort. * marks low-power blocks (N/M < 10). ETF near zero indicates the Hessian bound is meaninglessly loose at D = 9 with zero-replaced compositions.

4. Bridge and Boundary Results

Pennsylvania (bridge, FAIL) fails only at $M = 12$ (1.092%). At $M = 2$ through $M = 6$, all decimation levels pass. This is the most informative result in EXP-02: it pinpoints monthly-to-annual compression as the stress threshold for a moderately concentrated state.

North Carolina (bridge, PASS) passes at all M despite higher compositional velocity (1.28) and higher zero-share rate than some interior states. Its passage suggests that velocity alone does not determine failure.

Boundary cohort fails clearly and progressively. Rhode Island (8.78% at $M = 12$) is dominated by gas at 93.7% average share. Delaware fails even at $M = 2$ (1.40%) due to extreme sparsity and velocity. These cases define the operating envelope boundary.

Pennsylvania — verdict: FAIL

M	blocks	delta_M	rel %	SNR	pass
2	150	-0.002334	0.208%	0.2234	PASS
3	100	-0.004086	0.364%	0.3194	PASS
4	75	-0.006653	0.592%	0.4504	PASS
6	50	-0.008001	0.712%	0.4423	PASS
12	25	-0.012274	1.092%	0.4797	FAIL

North Carolina — verdict: PASS

M	blocks	delta_M	rel %	SNR	pass
2	150	0.000482	0.039%	0.0263	PASS
3	100	-0.000610	0.050%	0.0272	PASS
4	75	-0.000244	0.020%	0.0094	PASS
6	50	-0.002096	0.172%	0.0661	PASS
12	25	-0.002606	0.213%	0.0581	PASS

5. Arithmetic Comparator: Geometric vs Arithmetic

Every entity was processed through both geometric-mean and arithmetic-mean decimation. The arithmetic-mean path serves as a diagnostic comparator only — it is not CoDa-compliant and is not part of the EITT instrument. Its purpose is to reveal the specific contribution of the geometric mean's log-space averaging.

Key finding: at $M = 12$, geometric-mean EITT passes for all interior states while arithmetic-mean EITT fails for California (1.38%) and Minnesota (1.41%). This confirms the geometric mean is the correct CoDa aggregator — it preserves entropy invariance where the arithmetic mean does not. However, the arithmetic comparator also reveals a universal pattern: the geometric mean always sees *more* entropy loss than the arithmetic mean. This observation leads directly to the linear contamination tuner.

State	Band	Geom delta	Geom rel%	Geom	Arith delta	Arith rel%	Arith
California	interior	0.012316	0.829%	PASS	0.020509	1.381%	FAIL
Minnesota	interior	0.006680	0.490%	PASS	0.019226	1.410%	FAIL
Texas	interior	0.002537	0.208%	PASS	0.008727	0.716%	PASS
Wisconsin	interior	-0.001307	0.111%	PASS	0.011174	0.946%	PASS
Pennsylvania	bridge	-0.012274	1.092%	FAIL	0.009793	0.872%	PASS
North Carolina	bridge	-0.002606	0.213%	PASS	0.015871	1.299%	FAIL
Rhode Island	boundary	-0.024459	8.784%	FAIL	0.011648	4.183%	FAIL
West Virginia	boundary	-0.015244	5.283%	FAIL	0.005806	2.012%	FAIL
Wyoming	boundary	-0.014435	2.824%	FAIL	0.009617	1.881%	FAIL
Delaware	boundary	-0.044750	6.441%	FAIL	0.052806	7.600%	FAIL

Table 4. Geometric vs arithmetic EITT at $M = 12$. Interior states: geometric passes, arithmetic fails for CA and MN. Pennsylvania: geometric fails, arithmetic passes — the crossover case.

6. The Linear Contamination Tuner (F17)

The most significant methodological finding in EXP-02 is the discovery that the geometric-arithmetic gap provides a built-in contamination meter. Define:

$$\text{gap}(M) = |\text{delta}_M^{\text{geom}} - \text{delta}_M^{\text{arith}}|$$

$$C_{\text{geom}}(M) = \text{gap}(M) / H_{\text{bar}}(1) \text{ [relative contamination]}$$

The gap grows monotonically with M for all 10 entities (50/50 measurements). Fitting $\text{gap}(M) = aM^2 + bM + c$ reveals that the quadratic coefficient a is **negative** for every entity, meaning the gap grows linearly with saturation. The dominant term is the linear coefficient b. R^2 exceeds 0.96 in all cases.

This means the decomposition engine has a built-in governor: the geometric mean's log-space stress is self-limiting, not runaway. The tool saturates before it explodes.

The signed gap is negative for all 50 measurements: geometric decimation always sees *more* entropy loss than arithmetic. This is the Hessian effect made visible — the second-order curvature of entropy on the simplex systematically pulls the geometric mean toward the vertex.

Formal definition of F17:

F17 (Linear Contamination Tuner): $C_{\text{geom}}(M) = |\text{delta}_M^{\text{geom}} - \text{delta}_M^{\text{arith}}| / H_{\text{bar}}(1)$. Report for every tested M. The linear growth rate b of $C_{\text{geom}}(M)$ across M is the sensitivity dial. $C_{\text{geom}} < 0.5\%$ = clean. $C_{\text{geom}} 0.5\text{--}1\%$ = monitor. $C_{\text{geom}} > 1\%$ = investigate. $C_{\text{geom}} > 2\%$ = geometric mean is under material stress.

Critical constraint: F17 does NOT modify the EITT reading. It tells you when to distrust it. The tuner operates at the process-control layer, not the science layer. This preserves the gold-standard hierarchy: EITT is the science, the contamination meter is the honesty layer.

State	Band	C M=2	C M=3	C M=4	C M=6	C M=12	Max C
California	interior	0.119%	0.208%	0.256%	0.512%	0.552%	0.552%
Minnesota	interior	0.251%	0.432%	0.583%	0.792%	0.920%	0.920%
Texas	interior	0.137%	0.203%	0.334%	0.416%	0.508%	0.508%
Wisconsin	interior	0.291%	0.464%	0.651%	0.895%	1.057%	1.057%
Pennsylvania	bridge	0.406%	0.749%	1.105%	1.376%	1.964%	1.964%
North Carolina	bridge	0.355%	0.646%	0.815%	1.211%	1.512%	1.512%
Rhode Island	boundary	4.055%	5.925%	8.286%	10.465%	12.967%	12.967%
West Virginia	boundary	1.764%	2.974%	4.049%	5.494%	7.295%	7.295%
Wyoming	boundary	0.769%	1.091%	1.981%	4.031%	4.705%	4.705%
Delaware	boundary	3.315%	6.996%	9.131%	10.911%	14.041%	14.041%

Table 5. Relative contamination $C_{\text{geom}}(M) = \text{gap}(M) / H_{\text{bar}}(1)$ for all entities. Green < 0.5%, yellow 0.5–1%, orange 1–2%, red > 2%.

7. Pennsylvania Deep Dive: The Orthogonal Peak

Pennsylvania is the first case where geometric EITT fails (1.092%) while arithmetic EITT passes (0.872%) at the same M. The deltas move in *opposite directions*: geometric delta = -0.01227 (entropy drops) vs arithmetic delta = +0.00979 (entropy rises). This crossover reveals the mechanism.

At M = 12, an entire annual cycle is compressed into a single block. If any month has a near-zero carrier, $\log(10^{-6}) = -13.8$ pulls the geometric mean hard toward the simplex vertex. The arithmetic mean simply averages the small value linearly and barely notices. This is the 'fast sensitive peak' — a single month's near-zero component acts as an orthogonal impulse in log-space that the geometric mean amplifies but the arithmetic mean ignores.

Pennsylvania's characteristics make it the ideal detector for this effect: zero-share rate 9.6%, mean velocity 1.36 (the highest in the bridge cohort), and a dominant-share average of 49.4%. The high velocity means the energy mix changes fast, and the fluctuating near-zero carriers create the log-space impulses.

Implication: The geometric-arithmetic crossover at M = 12 is not a failure of EITT theory. It is a success of the contamination meter. $C_{\text{geom}} = 1.964\%$ at M = 12 for Pennsylvania correctly flags that the geometric mean is under material stress. The F17 tuner would have recommended investigation before the verdict was issued.

8. Stagewise RMSE and Fault Transparency

Per the process-control specification, every stage exposes local gauges and fault modes. The stagewise RMSE register for the interior cohort is summarized below.

Metric	California	Minnesota	Texas	Wisconsin
RMSE closure	1.14e-16	1.24e-16	1.32e-16	1.30e-16
RMSE dec_clr (mean)	1.45e-15	1.97e-15	2.35e-15	2.07e-15
RMSE phase (mean)	0.001370	6.01e-04	2.76e-04	3.40e-04
RMSE phase (max)	0.003051	0.001262	5.68e-04	5.77e-04
RMS delta_H (F13)	0.007081	0.004431	0.001462	9.69e-04
RMS rel (F14)	0.004768	0.003249	0.001199	8.20e-04
RMS SNR (F15)	0.331019	0.126686	0.078034	0.034211
RMS ETF (F16)	8.64e-09	3.52e-09	1.36e-09	9.75e-10

Table 6. Stagewise RMSE register for interior cohort.

Common faults across all entities:

S4 CAUTION — delta below 0.01: Every entity has delta_min at the zero-replacement floor (10^{-6}). This is the binding fault for the entire EXP-02 run. It means the Hessian bound is meaninglessly loose (ETF near 0) and Gamma is extremely large. The delta altimeter reads at the tool floor, not the data floor. Zero-handling is the dominant contamination source.

S8 INFO — sub-noise-floor: All EITT deltas are below the statistical noise floor (max SNR < 1.3 across all entities). The tool detects invariance but cannot statistically distinguish signal from noise. Binding constraint: detectability for interior cases, threshold for boundary cases.

9. Operating Envelope and Companion Diagnostics

State	Band	Gamma (M=2)	Binding	Hard flags	Soft flags
California	interior	1318934	detectability	none	delta < 0.01 (caution); EITT relative > 0.5% (inve
Minnesota	interior	1822512	detectability	none	delta < 0.01 (caution); Gamma > 100: 1822511.9
Texas	interior	1737693	detectability	none	delta < 0.01 (caution); Gamma > 100: 1737692.8
Wisconsin	interior	1388157	detectability	none	delta < 0.01 (caution); Gamma > 100: 1388156.8
Pennsylvania	bridge	1635993	EITT_threshold	none	delta < 0.01 (caution); EITT relative > 0.5% (inve
North Carolina	bridge	2606342	detectability	none	delta < 0.01 (caution); Gamma > 100: 2606342.2
Rhode Island	boundary	4817095	EITT_threshold	none	delta < 0.01 (caution); EITT relative > 0.5% (inve
West Virginia	boundary	1513491	EITT_threshold	none	delta < 0.01 (caution); EITT relative > 0.5% (inve
Wyoming	boundary	1078225	EITT_threshold	none	delta < 0.01 (caution); EITT relative > 0.5% (inve
Delaware	boundary	5096980	EITT_threshold	none	delta < 0.01 (caution); EITT relative > 0.5% (inve

Table 7. Operating envelope summary.

Companion diagnostics are reported separately per the four-instrument architecture: EITT (thermometer), velocity (compass), delta (altimeter), trajectory efficiency (GPS). The velocity scope shows Pennsylvania and Delaware as the highest-motion states, while trajectory efficiency is below 0.07 for all entities, indicating oscillatory rather than directional compositional change.

10. HIVP Regression Check

The Higgins Iterative Validation Protocol requires that every modification be regressed through all prior DUTs before advancement. EXP-02 introduces one new candidate formula:

F17 (Linear Contamination Tuner): measures the geometric-arithmetic gap as a process-control signal. This formula does not modify the core EITT observable (F2–F4). It adds a monitoring instrument at the process-control layer.

Regression status:

DUT	D	N	Core EITT	F17 status	Chain
EXP-01 Gold/Silver	2	624	PASS (0.60%)	Pending: re-run with F17	EXP-01 passed
EXP-02 Interior	9	300	PASS (max 0.83%)	C_geom < 1.06%	EXP-01 -> EXP-02 passed
EXP-02 Bridge (PA)	9	300	FAIL (1.09%)	C_geom = 1.96% (flagged)	Bridge case, expected
EXP-02 Boundary	9	300	FAIL (2.8-8.8%)	C_geom = 4.7-14.0% (flagged)	Boundary case, expected

Table 8. HIVP regression chain status. F17 is candidate — must be regressed through EXP-01.

No modifications to F1–F12 were required for EXP-02. The same pipeline that passed EXP-01 at D = 2 passes EXP-02 at D = 9 for interior compositions. F17 is a new addition that must be confirmed on EXP-01 before promotion to canonical status. The re-run of EXP-01 with F17 is the next mandatory step.

11. Conclusions and Next Steps

Primary conclusions:

1. EITT passes at $D = 9$ for diversified US energy compositions. The 1% threshold holds for all interior states across all tested decimation levels ($M = 2$ through 12). The HIVP chain extends from EXP-01 ($D = 2$) to EXP-02 ($D = 9$) without modification to the core formulas.
2. The geometric mean is confirmed as the correct CoDa aggregator. It preserves entropy invariance where the arithmetic mean fails (California and Minnesota at $M = 12$).
3. The geometric-arithmetic gap provides a built-in contamination meter (F17). The gap grows linearly with saturation in M , meaning the tool has a self-limiting governor. C_{geom} cleanly separates interior ($< 1.06\%$), bridge ($1.5\text{--}2.0\%$), and boundary ($> 4.7\%$) behavior.
4. Pennsylvania demonstrates that the geometric mean can amplify near-zero log-space impulses at high compression. F17 correctly flags this as material stress before the verdict is issued.
5. Zero-handling remains the dominant contamination source at $D = 9$. Δ_{min} sits at the replacement floor for all entities, rendering the Hessian bound uninformative.

Next steps:

1. Re-run EXP-01 gold/silver with F17 to confirm the tuner on the baseline.
2. Run the Europe monthly cohort (Spain, Germany, Italy, UK).
3. Investigate subcompositional EITT to address the zero-dominance problem.
4. Begin bridge-case deep dives for operating-envelope calibration.

Appendix A: Full Formula Catalog (F1–F17)

ID	Name	Formula
F1	Composition	$x_i(t) = f(R_i(t)) / \sum_j f(R_j(t))$, default $f(r) = r$
F2	Shannon entropy	$H(t) = -\sum_i x_i(t) \ln(x_i(t))$
F3	Geometric-mean decimation	$g_i(k) = [\prod x_i(t)]^{(1/M)}$, re-close
F4	EITT delta	$\delta_M = H_{\text{bar}}(M) - H_{\text{bar}}(1)$, $\text{relative}_M = \delta_M /H_{\text{bar}}(1)$
F5	Hessian bound	$ \delta_M \leq (D-1) \sigma_A^2 / (2 \delta M)$
F6	CLR transform	$\text{clr}_i = \ln(x_i) - (1/D) \sum_j \ln(x_j)$
F7	Aitchison variance	$\sigma_A^2 = (1/D) \sum_i \text{var}(\text{clr}_i)$
F8	Velocity	$v(t) = \ \text{clr}(t) - \text{clr}(t-1)\ _2$
F9	Vertex distance	$\delta(t) = \min_i x_i(t)$
F10	Trajectory efficiency	$\eta = \text{net} / \text{path} = \ \text{clr}(T) - \text{clr}(1)\ / \sum v(t)$
F11	Noise floor	$\text{SE}(M) = \sigma_H / \sqrt{N/M}$
F12	Complexity	$\Gamma = (D-1) \sigma_A^2 / (\delta \sqrt{N/M})$
F13	RMS abs error	$\text{RMS}_{\text{dH}} = \sqrt{(1/K) \sum \delta_M^2}$
F14	RMS rel error	$\text{RMS}_{\text{rel}} = \sqrt{(1/K) \sum (\delta_M/H_{\text{bar}})^2}$
F15	RMS noise-norm	$\text{RMS}_{\text{SNR}} = \sqrt{(1/K) \sum (\delta_M /\text{SE})^2}$
F16	RMS bound-util	$\text{RMS}_{\text{ETF}} = \sqrt{(1/K) \sum (\delta_M /B)^2}$
F17	Linear contam. tuner	$C_{\text{geom}}(M) = \delta_M^{\text{geom}} - \delta_M^{\text{arith}} / H_{\text{bar}}(1)$

Table A1. Complete EITT formula catalog. F1–F12 canonical (EXP-01). F13–F16 proposed (spec addendum). F17 candidate (EXP-02, pending HIVP regression on EXP-01).

Gold standard: Make EITT the science. Make HIVP the method of development. Make the operating envelope the honesty layer. Make HUF-GOV the application family.